Physical-Time Action Relaxation in an Elastic Collision Bath

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Abstract

An elastic repeated-collision model produces an action-valued covariance observable that relaxes exponentially in physical time to a positive constant. We derive its rate and plateau directly from the masses, bath velocity variance and collision frequency. Bounded bath velocities keep every collision below a fixed speed ceiling. The exact parameter dependence isolates the additional law needed for a universal action scale: energy and momentum conservation determine the collision map, while the reservoir statistics set the plateau. An equal-mass two-speed gas supplies a spatial realization: independent Poisson spacings generate reversal rate ρu and plateau mu/ρ.

1 A momentum receiver and a classical stochastic clock

A tracer of mass $m \geq M > 0$ meets fresh bath particles of mass M on the line. Collisions occur at an independent Poisson rate $\nu > 0$. Incoming bath velocities W_n are independent, identically distributed, centered and obey $|W_n| \leq w < c/3$, with variance $s^2 > 0$. Initial tracer velocity is independent of the bath and clock, centered, and bounded by w. Between collisions it is constant; $X(t) = X(0) + \int_0^t V(r)\, dr$.

The nontrivial one-dimensional elastic collision map is

$$V' = aV + bW, \quad W' = \frac{2m}{m+M}V - aW, \qquad a = \frac{m-M}{m+M}, \quad b = \frac{2M}{m+M} = 1 - a. \tag{1}$$

Direct substitution gives $mV' + MW' = mV + MW$ and $m(V')^2 + M(W')^2 = mV^2 + MW^2$. Since $0 \leq a < 1$, the tracer update is a convex combination and preserves $|V| \leq w$. The departing particle obeys $|W'| \leq (3m - M)w/(m + M) < 3w < c$.

The model specifies a refreshed reservoir and a velocity-independent clock. These are Barkai's collision-model premises [3, pp. 3–5, (2)–(12)], specialized here to a bounded bath and $m \geq M$. Each collision accounts for momentum and energy transfer; the complete stream supplies fresh particles and removes departing ones. A spatial gas realization would additionally derive encounter rates and incoming velocity statistics. The speed ceiling here is a kinematic restriction on Newtonian collisions in the reservoir frame, rather than a Lorentz-covariant collision law.

2 Stationary covariance and the action plateau

Define the two inverse-time rates

$$\gamma = \nu(1-a) = \frac{2\nu M}{m+M}, \qquad \beta = \nu(1-a^2) = \frac{4\nu m M}{(m+M)^2}. \tag{2}$$

The velocity generator acts on bounded functions as $Lf(v) = \nu\{\mathbb{E}[f(av + bW)] - f(v)\}$.

Proposition 1 (Collision plateau)**.** *The stationary velocity law is unique on* $[-w,w]$ *and is represented by the convergent series* $V_* \stackrel{d}{=} b\sum_{j=0}^{\infty} a^j W_j$*. Its variance and autocovariance are*

$$S_* := \operatorname{Var} V_* = \frac{b^2 s^2}{1-a^2} = \frac{M}{m} s^2, \qquad C(r) = S_* e^{-\gamma r}. \tag{3}$$

Thus the stationary displacement observable $\mathsf{h}(\Delta) = m \operatorname{Var}[X(t+\Delta) - X(t)]/\Delta$ *satisfies*

$$\mathsf{h}(\Delta) = H_* \left[1 - \frac{1-e^{-\gamma\Delta}}{\gamma\Delta}\right], \qquad H_* := \frac{2mS_*}{\gamma} = \frac{(m+M)s^2}{\nu} > 0. \tag{4}$$

Proof. The series converges absolutely and stays in $[-w,w]$, since $b\sum a^j = 1$ (for $a = 0$ it is simply W_0). Separating its first term proves invariance under one collision. Coupling two initial velocities to the same clock and bath multiplies their difference by a^{N_t}. Its mean contraction factor is $\mathbb{E}a^{N_t} = e^{-\nu(1-a)t}$, also for $a = 0$ with $0^0 = 1$. Hence invariant probability laws on the bounded interval coincide. Independence and centering give the variance in (3). Conditioning on the collision count yields $\mathbb{E}[V(t+r) \mid V(t)] = e^{-\gamma r} V(t)$; stationary covariance follows. Integrating the covariance over the displacement square gives $\mathsf{h}(\Delta) = 2m\int_0^{\Delta}(1 - r/\Delta)C(r)\,dr$, which evaluates to (4). □

This proof uses a contracting affine collision update; it requires neither a finite velocity state space nor detailed balance of the refreshed bath. The energy-valued reservoir parameter $\Theta := Ms^2$ gives $mS_* = \Theta$ and $H_* = \Theta(m+M)/(\nu M)$. For a general bounded bath, Θ denotes twice the mean incoming kinetic energy, without assuming a thermodynamic equilibrium distribution. The stationary covariance and Green–Kubo integral also follow from Barbier and Trizac's field-free Maxwell-kernel specialization [2, p. 19, (89)–(98)]. Their kernel exponent called ν is zero there; our ν is a positive collision frequency.

3 An observable relaxing in physical time

The integrated future-lag covariance defines a second operational quantity:

$$\mathsf{a}(t) := 2m \int_0^{\infty} \operatorname{Cov}[V(t+r), V(t)]\,dr. \tag{5}$$

It has action units and uses physical preparation time t, with the lag integrated out. Its definition differs from the finite-duration displacement observable; both share the same stationary long-duration plateau.

Proposition 2 (Physical-time relaxation)**.** *For the centered initial ensemble specified above, with* $S_0 = \operatorname{Var} V(0)$*,*

$$S(t) = S_* + (S_0 - S_*)e^{-\beta t}, \tag{6}$$

$$\mathsf{a}(t) = \frac{2mS(t)}{\gamma} = H_* + (\mathsf{a}(0) - H_*)e^{-\beta t}, \qquad \dot{\mathsf{a}} = \beta(H_* - \mathsf{a}). \tag{7}$$

In particular a tracer prepared at rest has $\mathsf{a}(t) = H_*(1 - e^{-\beta t}) > 0$ *for every* $t > 0$ *and* $\mathsf{a}(t) \to H_* > 0$ *as* $t \to \infty$*.*

Proof. The generator gives $Lv = -\gamma v$ and $L(v^2) = -\beta v^2 + \nu b^2 s^2$. Centering is preserved, so $\dot{S} = -\beta S + \nu b^2 s^2 = \beta(S_* - S)$. The conditional mean from Proposition 1 gives, even away from stationarity, $\operatorname{Cov}[V(t+r), V(t)] = e^{-\gamma r} S(t)$. Integration proves (7). □

The original displacement definition retains a distinct duration parameter:

$$\mathsf{h}_\Delta(t) = \frac{2m}{\gamma\Delta}\int_0^\Delta S(t+r)\left[1 - e^{-\gamma(\Delta-r)}\right] dr. \tag{8}$$

Indeed, for $r \le s$ the covariance inside its double integral is $e^{-\gamma(s-r)}S(t+r)$; integrating over s proves the formula. At fixed Δ, it approaches the stationary value in (4) exponentially at rate β. At every fixed t it tends to zero as $\Delta \downarrow 0$, with $\mathsf{h}_\Delta(t) \sim mS(t)\Delta$ when $S(t) > 0$. Its long-duration limit is H_*, since the transient contribution to (8) is bounded in magnitude by $2m|S_0 - S_*|e^{-\beta t}/(\gamma\beta\Delta)$.

4 What would select a common constant?

For fixed bath mass, variance and clock, the plateau grows as $m + M$. Achieving the same H_* for different tracer masses requires

$$\nu_m = \frac{(m+M)s_m^2}{H_*}. \tag{9}$$

This equation is a diagnostic of a proposed encounter law. Binary energy and momentum conservation alone place no restriction of this form on the external clock or reservoir preparation.

More explicitly, take a fixed admissible bath and a tracer initially at rest. For any $0 < \epsilon \le 1$, replace every incoming velocity by $W_n^{(\epsilon)} = \epsilon W_n$, keeping masses and clock unchanged. Every elastic collision, centering assumption and speed bound remains valid. Both H_* and $\mathsf{a}(t)$ are multiplied by ϵ^2. Consequently this entire class admits arbitrarily small positive plateaus even though every nondegenerate member relaxes to its own positive constant. This is a countermodel test for these specified stochastic collision premises.

The next model should replace the supplied clock by a spatial encounter law and test how it scales with reservoir velocity and density. The physical-time relaxation mechanism is already explicit: persistent injection of velocity variance balances collisional contraction. A universal quantum action scale requires an additional physical relation that fixes this balance across systems.

Ben-Naim and Krapivsky provide a useful clock comparison: exponential decay in collision number becomes algebraic decay in physical time in their cooling model [4, p. 6, (14)–(17)].

5 A spatial receiver with an exact collision clock

An infinite equal-mass hard-point gas realizes the two-speed action plateau with every particle's speed bounded by $u < c$. Its encounter clock follows from the initial spatial statistics. This is the dichotomic Jepsen-gas case analyzed by Balakrishnan, Bena and Van den Broeck [1, pp. 7–8, (11)–(12), pp. 17–19]. The following construction makes its clock explicit.

Fix $m, \rho, u > 0$. Put a tagged particle at the origin and independently put Poisson particles of intensity ρ on the rest of the line. Give every particle an independent sign, with velocities $+u$ and $-u$ equally likely. This is the Palm preparation: a typical particle is placed at zero, with independent gas on either side. Particles move freely between instantaneous elastic collisions, exchanging velocities and preserving their spatial order. Each collision conserves momentum and kinetic energy. The initial energy per unit length has mean $\rho m u^2/2$; the reservoir has infinite extent.

The equivalent free-trajectory construction lets particles cross and exchanges their physical labels at crossings. Each marked Poisson stream is translated rigidly, so the untagged homogeneous marked-gas law is invariant. A tagged path on $[0,T]$ lies inside $[-uT,uT]$; only free lines starting in $[-2uT,2uT]$ can intersect that region during this interval. There are almost surely finitely many such lines and crossings. This constructs the local dynamics without an accumulation of collisions. Positions are continuous and piecewise linear; velocities are right-continuous with impulsive changes.

Proposition 3 (Spatially generated reversals)**.** *In this preparation the tagged velocity is a stationary symmetric two-state Markov chain with reversal rate $\lambda = \rho u$. Consequently*

$$C(s) = u^2 e^{-2\rho u s}, \tag{10}$$

$$\mathsf{h}(\Delta) = \frac{mu}{\rho}\left[1 - \frac{1-e^{-2\rho u\Delta}}{2\rho u\Delta}\right], \qquad H_* = \frac{mu}{\rho} > 0. \tag{11}$$

Here $\mathsf{h}(\Delta) = m\,\mathrm{Var}[X(t+\Delta)-X(t)]/\Delta$, and the limit is $\Delta \to \infty$ at fixed m, ρ, u in the infinite gas.

Proof. Condition first on tagged initial velocity $+u$. Write $B_1 < B_2 < \cdots$ for the initial positions of negative-velocity free lines to its right, and $-A_1 > -A_2 > \cdots$ for positive-velocity lines to its left; set $A_0 = B_0 = 0$. Independent thinning makes (A_j) and (B_j) independent Poisson arrival sequences of intensity $\rho/2$. Their successive gaps are independent exponentials of that rate.

The tag first switches from the line ut to $B_1 - ut$, then to $-A_1 + ut$, then to $B_2 - ut$, and so on. Lines with the same velocity remain parallel; at each switch the next opposing line is the next unused member of the corresponding ordered sequence. Solving these intersections gives

$$t_{2j-1} = \frac{B_j + A_{j-1}}{2u}, \qquad t_{2j} = \frac{B_j + A_j}{2u}.$$

Thus successive waiting times alternate between $(B_j - B_{j-1})/(2u)$ and $(A_j - A_{j-1})/(2u)$: all are independent exponentials of rate ρu. Reflection proves the same law for initial velocity $-u$. The independent uniform initial sign is the stationary law of this chain. Its covariance follows by taking the parity of its Poisson reversal count. Integrating $2m\int_0^\Delta (1-s/\Delta)C(s)\,ds$ proves the action formula. □

The mechanical statement agrees with the source's covariance (12). Its discussion distinguishes a tag initially at $\pm u$ from a tag initially at rest: p. 19 gives the ballistic term $e^{-\rho u t}\delta(X \mp ut)$ in the former case, and a Bessel-weighted atom at zero in the latter. Our proposition uses the former preparation throughout.

5.1 The scale supplied by spatial preparation

The mean collision duration is $1/(\rho u)$, and the plateau has the action units of momentum times mean spacing: $H_* = mu/\rho$. At fixed density, cooling $u \mapsto \epsilon u$ changes the collision rate by ϵ and the plateau by ϵ, rather than the quadratic scaling of the independently clocked bath. At fixed u, increasing ρ also closes the plateau. Each family retains the hard speed ceiling. A shared value K across these gases therefore requires $\rho_m = mu_m/K$; a preparation law would have to select this relation.

This realization advances the mechanical-clock obligation and inherits the two-speed bridge and cut results through equality of the tagged path laws. The positive plateau concerns long observation windows; microscopic refinement still gives $\mathsf{h}(\Delta) \sim mu^2\Delta \to 0$. The next test holds m, u, ρ fixed while changing spacing correlations, to identify how much of the plateau is supplied by Poisson preparation. This is a classical impulsive gas in a chosen frame, with a kinematic speed ceiling; a relativistic collision theory or a smooth-force realization would add further dynamics.

6 Return to the partition question

The project's central question concerns a sequence of time partitions $\pi_N = \{0 = t_0^{(N)} < \cdots < t_N^{(N)} = T\}$ with mesh tending to zero. The present relaxation law describes preparation time under a supplied bath. It therefore serves as a comparison mechanism for the partition problem: one must derive, from admissible cut-point refinements, both the quantity that survives and the condition selecting its positive value. The collision calculation isolates how a positive constant can instead enter through reservoir energy and a clock. The next partition test tracks these inputs explicitly when a refinement law is proposed.

References

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